# Day 21: Model comparison for logistic regression

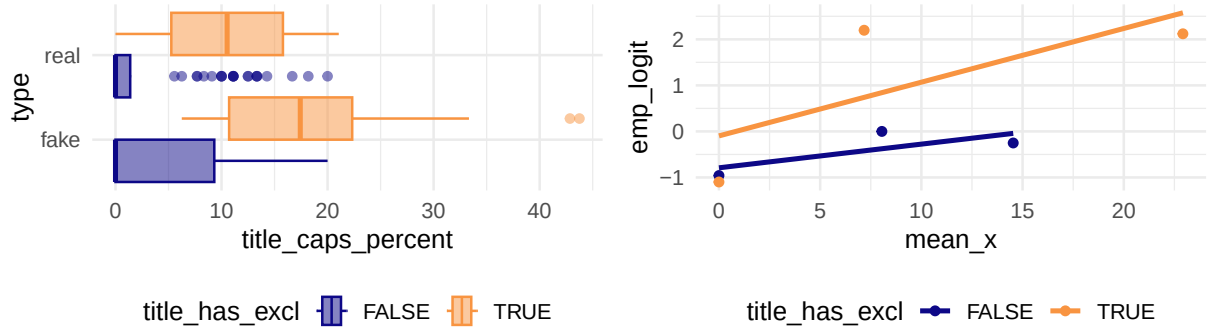
Last time we learned how to do inference for **one** coefficient (Wald test and CI) and how to check conditions. Today we compare **models**:

1. Do we need an interaction term?
2. The **likelihood ratio test** (LRT): the logistic version of the extra sums of squares $F$-test
3. AIC and BIC for logistic regression

## The model from last time

| term | estimate | std. error | z | p-value |
|---|---|---|---|---|
| Intercept | -2.232 | 0.672 | -3.32 | <0.001 |
| title_words | 0.120 | 0.058 | 2.07 | 0.038 |
| title_has_exclTRUE | 1.998 | 0.849 | 2.35 | 0.019 |
| title_caps_percent | 0.050 | 0.029 | 1.69 | 0.091 |

**Question:** is the association between `title_caps_percent` and the odds of being fake news *different* for titles with and without an exclamation point? That is, do we need an **interaction** between `title_has_excl` and `title_caps_percent`?




## The model with an interaction

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{ words}_i + \beta_2 \text{ excl}_i + \beta_3 \text{ caps}_i + \beta_4 (\text{excl}_i \times \text{caps}_i)$$

| term | estimate | std. error | z | p-value |
|---|---|---|---|---|
| Intercept | -2.273 | 0.677 | -3.36 | <0.001 |
| title_words | 0.127 | 0.059 | 2.14 | 0.032 |

| term | estimate | std. error | z | p-value |
|---|---|---|---|---|
| title_has_exclTRUE | 1.281 | 1.386 | 0.92 | 0.355 |
| title_caps_percent | 0.041 | 0.033 | 1.28 | 0.202 |
| title_has_exclTRUE:title_caps_percent | 0.060 | 0.100 | 0.60 | 0.549 |

**Q1.** Write the fitted model on the log-odds scale for each group separately.

With an exclamation point ($excl = 1$):

Without an exclamation point ($excl = 0$):

**Q2.** Do we need an interaction term? Write out the null and alternative hypothesis.

**Q3.** Do we need `title_has_excl` at all? Write out the null and alternative hypothesis for this question.

## The likelihood ratio test (LRT)

In linear regression we compared nested models with the _____ $F$-test. For logistic regression we use a **likelihood ratio test**.

**Deviance**

The **deviance** of a model is $-2 \log L$, where $L$ is the maximized likelihood. Like SSE in OLS, a smaller deviance means a _____ fit. A bigger model always has a deviance that is the same or _____.

**Likelihood Ratio Test**

**Hypotheses:** $H_0$: _____ vs. $H_a$: at least one of these $\beta$'s $\neq 0$
**Test statistic:** $\Lambda$ = Dev(reduced) − Dev(full)
**Reference distribution:** $\chi^2$ with $df$ = _____
**p-value:** the area _____ $\Lambda$ (upper tail)

In R, give the **reduced** model first:

```
anova(reduced_model, full_model, test = "LRT")
```

| Resid DF | Resid Dev | Df | Deviance | Pr(>Chi) |
|---|---|---|---|---|
| $n-$ (p+1 in reduced) | Dev(reduced) | | | |
| $n-$ (p+1 in full) | Dev(full) | # extra coefs | $\Lambda$ | p-value |

## Example 1: do we need the interaction?

```
Analysis of Deviance Table

Model 1: type_num ~ title_words + title_has_excl + title_caps_percent
Model 2: type_num ~ title_words + title_has_excl * title_caps_percent
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       146     170.86
2       145     170.44  1  0.41625   0.5188
```

**Hypotheses:**

**Test statistic, df, and p-value:**

**Conclusion in context:**

### Example 2: the overall test

The summary() output gives the deviance of the model, and of an intercept-only model:

```
summary(news_mod)
```

```
Call:
glm(formula = type_num ~ title_words + title_has_excl + title_caps_percent,
    family = "binomial", data = news)

Coefficients:
                   Estimate Std. Error z value Pr(>|z|)
(Intercept)        -2.23184    0.67187  -3.322 0.000894 ***
title_words         0.12034    0.05808   2.072 0.038260 *
title_has_exclTRUE  1.99783    0.84918   2.353 0.018640 *
title_caps_percent  0.04982    0.02946   1.691 0.090783 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 201.90  on 149  degrees of freedom
Residual deviance: 170.86  on 146  degrees of freedom
AIC: 178.86

Number of Fisher Scoring iterations: 4
```

**Q4.** Fill in the ANOVA table below for the test $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ against $H_A$ at least $1 \neq 0$

| | Resid DF | Resid Dev | Df | Deviance | Pr(>Chi) |
|---|---|---|---|---|---|
| 1 | | | | | |
| 2 | | | | | |

```
1 - pchisq(lambda, df)
```

**Conclusion:**

### $z$-test vs. LRT

For one coefficient, $H_0 : \beta_i = 0$:

- In OLS, the $t$-test and $F$-test give *exactly* the same p-value.
- In logistic regression the $z$-test (Wald) and the LRT are _____ equivalent. Hopefully they are close. If not, the _____ is usually more accurate.

| term | z-test p-value | LRT p-value |
|---|---|---|
| title_words | 0.0383 | 0.0355 |
| title_has_excl | 0.0186 | 0.0075 |
| title_caps_percent | 0.0908 | 0.0869 |

The LRT can also test **several coefficients at once**, which the $z$-test cannot.

### Example 3: is `title_has_excl` needed at all?

`title_has_excl` appears in two terms of `news_mod2`. To drop it entirely we have to remove both the main effect and the interaction.

$H_0$: _____

```
Analysis of Deviance Table

Model 1: type_num ~ title_words + title_caps_percent
Model 2: type_num ~ title_words + title_has_excl * title_caps_percent
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       147     178.01
2       145     170.44  2   7.5695  0.02271 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Q5.** The individual $z$-tests for these two terms have large p-values. The joint LRT has a small p-value. Why? (Think back to the $F$-test in MLR.)

**Conclusion in context:**

## AIC and BIC for logistic regression

$$AIC = -2\log(L) + 2k \qquad BIC = -2\log(L) + \log(n) \cdot k$$

where $L$ is the maximized likelihood and $k$ is the number of coefficients (including the intercept). For binary logistic regression, $-2\log(L)$ is the _____. Smaller is better, and both can compare nested **or** non-nested models.

| | df | AIC | BIC |
|---|---|---|---|
| reduced_model | 3 | 184.01 | 193.04 |
| news_mod | 4 | 178.86 | 190.90 |
| news_mod2 | 5 | 180.44 | 195.50 |

**Q6.** Which model do AIC and BIC prefer? Do they agree with the LRTs?

## Summary

| To compare… | Use |
|---|---|
| one coefficient to 0 | |
| several coefficients to 0, nested models | |
| non-nested models, or many models | |

- Deviance is $-2\log L$. It plays the role of _____ in OLS.
- We use deviance to compare models

## R function reference

| Function | Purpose |
|---|---|
| `anova(reduced, full, test = "LRT")` | Likelihood ratio test for nested models |
| `deviance(model)` | Deviance ($-2\log L$) of a model |
| `pchisq(x, df)` | $\chi^2$ probability; use `1 - pchisq()` for a p-value |
| `AIC(m1, m2, ...)`, `BIC(m1, m2, ...)` | Information criteria for one or more models |